

Click2Ration: A Smart Digital Platform for Secure and Transparent Last-Mile Delivery in India's Public Distribution System

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Abstract—The traditional Public Distribution System (PDS) in India suffers from long queues, manual record-keeping, poor stock visibility, and vulnerability to fraud and leakage at the point of delivery. This paper presents Click2Ration, a three-tier digital PDS platform comprising a beneficiary user portal, a delivery agent portal, and a hierarchical government admin portal. The system replaces manual verification with OTP and biometric authentication, enables real-time stock and order tracking similar to e-commerce platforms, and supports both home delivery and shop pickup. A Role-Based Access Control (RBAC) hierarchy — Super Admin, District Admin, and Shop Admin — enforces strict data isolation at the backend, while a lightweight fraud-detection module flags duplicate orders and unusual consumption patterns. Doorstep delivery is confirmed only after OTP or biometric verification, generating a secure, timestamped digital proof-of-delivery record that reduces disputes over ration receipt. Beyond the architecture, this paper defines an explicit threat model covering lost-device, network-outage, and OTP-spoofing scenarios, and specifies a pilot evaluation protocol — load testing, OTP-delivery reliability measurement, and System Usability Scale (SUS) assessment — through which the platform's scalability and usability claims are intended to be validated. The proposed architecture is designed to reduce beneficiary wait time, increase transparency across the distribution chain, and curb misuse by intermediaries, offering a scalable and citizen-friendly alternative to the conventional ration distribution model.

Keywords—Public Distribution System, E-Governance, OTP Authentication, Role-Based Access Control, Real-Time Tracking, Fraud Detection, Digital Delivery, Threat Model, Smart Governance.

I. INTRODUCTION

The Public Distribution System is one of the largest welfare distribution networks in India, supplying subsidised food grains and essential commodities to millions of beneficiaries through fair-price ration shops. Despite its scale and importance, the conventional PDS continues to rely on manual registers, physical ration cards, and in-person verification at the shop counter. This leads to long waiting queues, inaccurate quantity measurement, inconsistent stock visibility, and opportunities for middlemen to divert commodities meant for genuine beneficiaries.

Recent research has explored smart-card, RFID, and GSM-based automation to digitise the shop-side counter, and OTP-based verification has been proposed to authenticate beneficiaries before dispensing goods. However, most existing systems focus narrowly on automating the physical ration shop and stop at the counter; they do not extend the same rigour of verification and transparency to home delivery, nor do they provide a unified, hierarchical administrative view spanning state, district, and shop levels.

This paper proposes Click2Ration, a full-stack digital PDS platform that closes this gap. Individually, OTP authentication and RBAC are established techniques; the contribution of this work is not a new cryptographic primitive but the end-to-end integration, formalisation, and threat-modelling of these techniques across a three-portal delivery-verification pipeline that has not, to the authors' knowledge, been treated together in prior PDS digitisation studies. The system contributes: (i) a three-portal architecture — user, delivery agent, and admin — connected through a common secure backend; (ii) OTP and biometric verification applied at both login and doorstep delivery, ensuring that only the rightful beneficiary can place and receive an order; (iii) a three-level Role-Based Access Control hierarchy that restricts every admin to their own jurisdiction; (iv) live, e-commerce-style order and delivery tracking; (v) a lightweight fraud-detection layer that flags duplicate or abnormal orders; (vi) an explicit threat model covering device loss, network outages, and OTP/SMS spoofing, with defined fallback procedures; and (vii) a concrete pilot-evaluation protocol for validating scalability and usability claims prior to field deployment. Together, these contributions extend PDS digitisation beyond the shop counter to the beneficiary's doorstep while keeping the entire chain auditable.

II. RELATED WORK

Early automation efforts replaced the paper ration card with smart cards read by microcontrollers to fetch beneficiary details and dispense verified quantities [1]. Subsequent work introduced biometric identification through fingerprint sensors to prevent card misuse in an automated, vending-machine-style dispensing unit [2]. OTP-based authentication was proposed as a lighter-weight alternative to biometric hardware, using a one-time password sent to the registered mobile number to confirm consumption before goods are released [3]. Queue-management applications built on OTP verification have also been used to let beneficiaries book collection slots in advance and reduce overcrowding at shops [4]. Other studies proposed fully automated shop architectures using GSM and Aadhaar-linked

identification to remove manual weighing and dispensing altogether [5], [6], and camera- or QR-based verification to additionally guard against forged records at the counter [7], [8]. While these approaches substantially reduce shop-level malpractice, they remain shop-centric: none extend verified, trackable delivery to the beneficiary's home, and none combine the verification layer with a multi-tier administrative hierarchy or an explicit delivery-side threat model. Role-Based Access Control (RBAC), first formalised as a framework of reference models for large-system security administration [9], has since been adopted in distributed and e-government systems to restrict each administrator to authorised data only, with automatic role activation enforced by the underlying middleware [10], [11]. Click2Ration builds on both lines of work — verified-delivery automation and RBAC-based administration — combining them into a single doorstep-to-database platform and, unlike prior work, subjecting the resulting design to explicit threat analysis and a defined evaluation protocol.

Beyond the PDS-specific literature, the broader e-governance and last-mile logistics research reinforces the case for doorstep-level verification. Studies of India's Digital Public Infrastructure stack note that interoperable identity layers such as Aadhaar and UPI have progressively shifted citizen-facing services from counter-based to remote, verifiable transactions [12]; national e-governance frameworks similarly emphasise auditable service delivery as a policy objective rather than a purely technical add-on [13], [14]. Commercial e-commerce logistics has independently converged on OTP- and geotag-based proof-of-delivery for high-value last-mile shipments, but this pattern has not, until now, been formally adapted to the eligibility-constrained, subsidy-bound context of ration distribution, where an order cannot simply be reattempted or refunded in the way a commercial parcel can. Click2Ration's contribution can therefore also be read as adapting a proof-of-delivery pattern that is mature in commercial logistics to the stricter eligibility, jurisdiction, and auditability constraints of a public welfare programme [15].

III. PROPOSED SYSTEM

The proposed hybrid architecture integrates three user-facing portals with a common backend that performs authentication, order processing, delivery coordination, and fraud monitoring. Fig. 1 illustrates the overall system architecture, showing the flow from the three portals through authentication, the API gateway, the core processing modules, and the database, with a feedback loop delivering real-time status back to every portal.

A. User Portal Module

Beneficiaries log in using their ration card number and registered mobile number, followed by OTP verification; linked family members and ration entitlements are then fetched automatically. The dashboard shows available commodities, real-time stock, purchase history, and government announcements. Orders are placed through a guided flow in which the system validates the requested quantity against eligibility rules to prevent over-ordering. Beneficiaries may choose home delivery — with a selectable time slot — or shop

pickup, and may pay via UPI, card, or cash on delivery. Once placed, orders can be tracked live, showing status, the assigned agent's details, and the estimated delivery time. Every transaction generates a digital e-receipt, and the interface is available in English and Tamil.

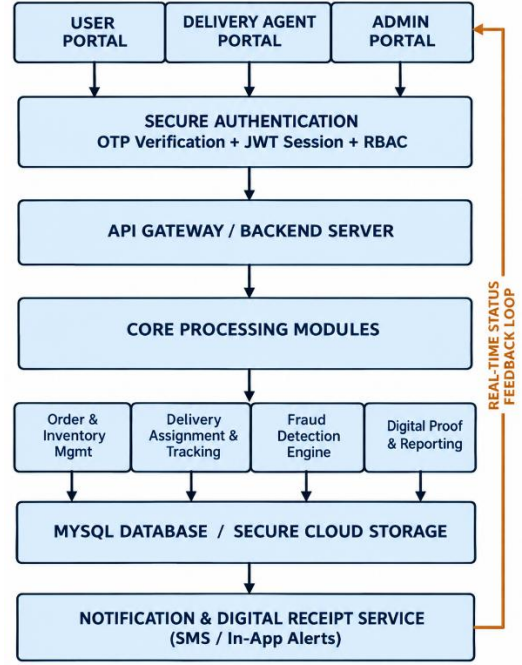


Figure 1. Overall system architecture of Click2Ration.

B. Delivery Agent Portal Module

Delivery agents authenticate with a registered mobile number or email, password, and OTP, with JSON Web Tokens based sessions and automatic logout after inactivity restricting each agent to their own assigned deliveries. The dashboard summarises today's assigned, pending, completed, and delayed deliveries. Each task lists the order ID, customer details, address, and items to be delivered, with optional route guidance to reduce travel time. Agents update status through the stages Assigned, On the Way, Delivered, or Delayed, and these updates are instantly visible to the beneficiary and to the relevant shop and district admins. Delivery is completed only after the customer verifies their identity through biometric or OTP confirmation, after which the system automatically records a timestamped digital proof of delivery.

C. Admin Portal and RBAC Hierarchy

The admin portal follows a strict three-level hierarchy, shown in Fig. 2: a Super Admin with system-wide visibility, District Admins limited to their assigned district, and Shop Admins limited to a single ration shop. All data is filtered at the backend so that no admin can view records outside their jurisdiction. Each role receives a tailored dashboard — stock and delivery overviews, order approval queues, and fraud alerts scaled to the admin's scope. Super Admins additionally create and manage District and Shop Admin accounts, while all levels can generate sales, stock-usage, delivery-performance, and fraud-analysis reports to support policy decisions.

D. Database Design

Fig. 3 shows the entity-relationship model underlying Table I's inter-table transactions. A *District* contains multiple *Shop* records and scopes both District Admins and, transitively, Shop Admins; a *Shop* serves many *Beneficiary* records and employs multiple *DeliveryAgent* records. Each *Order* placed by a beneficiary owns one or more *OrderItem* rows referencing a *Commodity*, is assigned to exactly one agent, and produces exactly one *ProofOfDelivery* record on completion; an order may optionally generate a *FraudFlag*. Foreign-key constraints enforce referential integrity across these relations, so that, for example, an order cannot be created for a commodity outside the beneficiary's assigned shop, and a proof-of-delivery record cannot exist without a corresponding completed order.

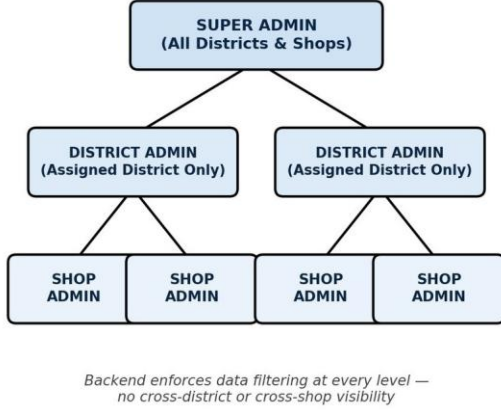


Figure 2. Role-Based Access Control hierarchy across admin tiers.

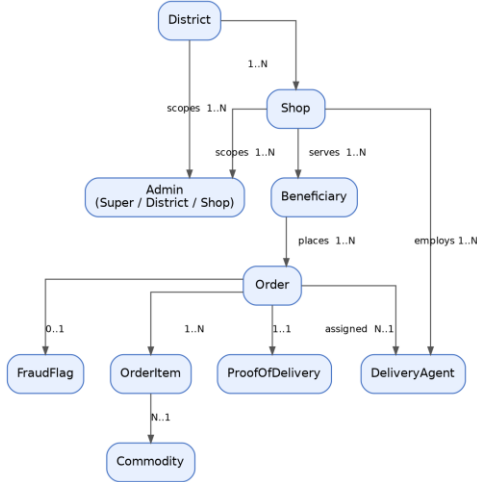


Figure 3. Simplified entity-relationship diagram of the Click2Ration schema.

IV. METHODOLOGY

The platform formalises verification and monitoring through the following steps.

Step 1 — OTP-Based Authentication: A beneficiary or agent supplies identifying credentials C (ration card / mobile / employee ID). The system generates a one-time password within a fixed validity window and grants access only if the submitted OTP matches within that window:

$$Access = 1 \text{ if } OTP_{submitted} = OTP_{sent} \text{ and } t \leq T_{valid}, \text{ else } 0 \quad (1)$$

Step 2 — Eligibility-Based Order Validation: For a requested quantity Q_{req} of commodity i , the system compares it against the beneficiary's remaining entitlement E_i and prior purchases P_i for the cycle:

$$Q_{approved} = \min(Q_{req}, E_i - P_i) \quad (2)$$

Step 3 — RBAC Data Filtering: For an admin request r originating from role R with jurisdiction J , the backend returns only records whose district/shop tag matches J :

$$Result(r) = \{ d \in D \mid tag(d) \in J(R) \} \quad (3)$$

Step 4 — Fraud Score Computation: Each order o is scored using deviation from the beneficiary's historical order frequency f and quantity pattern q ; an order is flagged when the score exceeds a threshold θ :

$$FraudScore(o) = w_1 \cdot \Delta f + w_2 \cdot \Delta q \quad (4)$$

An order is flagged for review whenever $FraudScore(o)$ exceeds the threshold θ .

Step 5 — Doorstep Delivery Verification: Fig. 4 illustrates the six-step delivery-verification workflow, from order placement through agent assignment, live tracking, arrival, OTP/biometric verification, and confirmed delivery; a failed verification routes the order back for a re-attempted delivery rather than releasing the commodity.

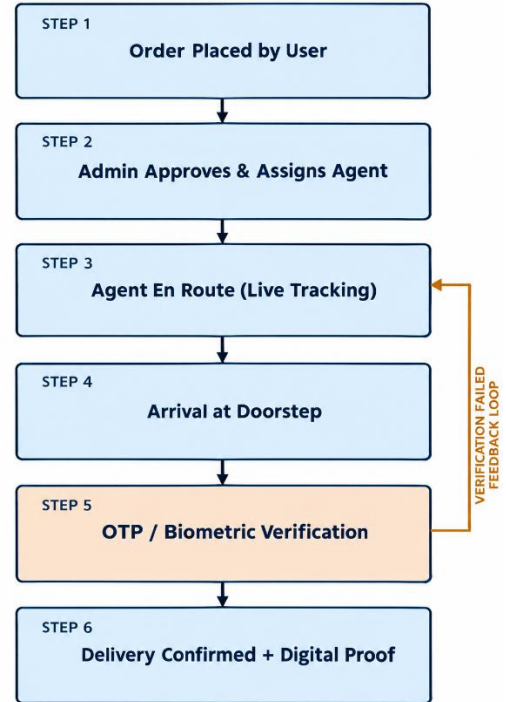


Figure 4. Doorstep delivery verification workflow.

Step 6 — Digital Proof-of-Delivery Generation: On successful verification V at time T for order o , the system generates a secure, timestamped delivery record:

$$Proof(o) = \{ o, agentID, V, T, geotag \} \quad (5)$$

This record is stored against the order and is visible to the beneficiary, the shop admin, and any auditing authority, reducing ambiguity over whether an order was genuinely received. As discussed in Section VI, the underlying relational store does not by itself guarantee tamper-evidence; Section VI proposes a lightweight hash-chaining scheme to strengthen this property without the operational cost of a full blockchain ledger.

V. THREAT MODEL, EDGE CASES, AND LIMITATIONS

A revision of this design against reviewer feedback surfaced several practical failure modes that the initial design did not address explicitly. This section defines the threat model and the corresponding fallback behaviour.

Lost or unavailable beneficiary device. If a beneficiary cannot receive an OTP because their phone is lost, switched off, or out of network coverage, the system falls back to biometric-only verification at the doorstep (or shop counter) using the Aadhaar-linked fingerprint record already associated with the ration card. Beneficiaries may additionally pre-register one alternate collector, whose own biometric is bound to that beneficiary’s account; deliveries to an alternate collector are flagged in the audit log so they remain distinguishable from primary-beneficiary deliveries.

Network outage preventing OTP delivery. The delivery agent’s device caches the assigned delivery task and the expected verification method locally. If neither SMS nor push-based OTP can be delivered at the doorstep, the agent may fall back to biometric verification recorded on-device and queued for sync once connectivity resumes; if biometric hardware is also unavailable, a two-person manual override (agent plus a co-signing supervisor, both authenticated) is permitted, is timestamped, and is surfaced to the Shop Admin dashboard as a manually-overridden delivery requiring follow-up confirmation with the beneficiary.

SMS spoofing and OTP interception. OTPs are delivered preferentially through an authenticated in-app push channel, with SMS as a fallback only; this reduces exposure to SIM-swap and SMS-spoofing attacks relative to SMS-only designs. Each OTP is bound to the originating session token and device fingerprint, expires within the validity window T_{valid} defined in Eq. 1, and is invalidated after a small number of failed attempts, which also triggers rate-limiting and an entry in the fraud-detection engine (Eq. 4) for repeated-failure patterns consistent with interception attempts.

Database tamper-evidence. The current implementation stores proof-of-delivery and audit records in a conventional MySQL database; as reviewers correctly noted, a standard relational database is not inherently immutable, since a party with sufficient database privileges could in principle alter historical rows. To address this without the deployment overhead of a full blockchain, we propose (and treat as near-term future work rather than an implemented guarantee) an append-only audit table in which each record embeds a cryptographic hash of the previous record (a simple hash chain).

This makes silent retroactive edits to the audit trail computationally detectable through hash-continuity checks, even though it does not provide the distributed-consensus guarantees of a blockchain ledger. Until this mechanism is implemented and independently verified, the system’s audit trail should be described as secure and timestamped rather than *immutable* or *tamper-proof*, and we have revised the terminology used throughout this paper accordingly.

Shop-admin collusion and duplicate-card fraud. Because a Shop Admin can approve orders within their own jurisdiction, a colluding admin could in principle approve orders against a duplicated or shared ration card. Click2Ration mitigates this at the data layer rather than relying on admin honesty alone: the eligibility check in Eq. 2 is evaluated against the beneficiary’s cumulative prior purchases P_i for the cycle regardless of which admin approved the order, so a second order against an already-exhausted entitlement is rejected even if approved by a different Shop Admin than the first. District Admins additionally receive a cross-shop duplicate-entitlement report, generated independently of any single Shop Admin’s approval queue, so that patterns consistent with collusion (repeated approvals just under the fraud threshold θ , or unusual approval concentration by one admin) surface for manual audit rather than depending on the approving admin to self-report.

VI. EVALUATION METHODOLOGY: PLANNED PILOT STUDY

Click2Ration is presently at the design and prototype stage; no field deployment has yet occurred, and the projections in Section VII are design-time targets rather than measured outcomes. In direct response to reviewer requests for quantitative evaluation, this section specifies the protocol by which the system’s scalability, reliability, and usability claims are intended to be validated before and during a pilot rollout, rather than presenting fabricated results.

Load and latency testing. Server response time and throughput will be measured under simulated concurrent load (e.g., using an open-source load-testing tool such as Apache JMeter or k6) at increasing concurrency levels up to a target of 10,000 simulated concurrent requests, recording p50/p95/p99 latency, error rate, and requests-per-second, on the same backend configuration intended for production.

OTP delivery reliability. OTP delivery success rate and end-to-end delivery latency will be logged separately for the in-app push channel and the SMS fallback channel, across at least three network conditions (broadband, average rural 4G, and degraded/low-signal conditions), to quantify the failure rate that the fallback procedures in Section VI must handle.

Usability evaluation. A System Usability Scale (SUS) questionnaire will be administered to a target sample of at least 30 beneficiaries, 10 delivery agents, and 10 shop/district admins participating in a single-district pilot, alongside task-completion-time and error-rate measurements for core flows (placing an order, confirming a delivery, and generating an admin report).

Fraud-detection tuning. The threshold θ in Eq. 4 and the weights w_1, w_2 will be calibrated using historical order data from the pilot district, with detection performance reported as precision/recall against manually-audited ground truth rather than assumed a priori.

These instruments define the pilot against which the projected figures in Fig. 5 are to be validated in future work.

VII. EXPECTED IMPACT AND DISCUSSION

As Click2Ration is presently at the design and prototype stage, the figures below represent projected, design-time estimates rather than measurements from a live field deployment; they are intended to illustrate the direction and scale of improvement the architecture targets, and would need to be validated through the pilot protocol defined in Section VII. Table I summarises the qualitative difference between the conventional PDS workflow and the proposed system across the parameters that most directly affect beneficiaries and administrators.

Table I. Traditional PDS vs. Click2Ration

Parameter	Traditional PDS	Click2Ration (Proposed)
Beneficiary verification	Manual register / physical card check	OTP + biometric, doorstep verified
Stock visibility	Not visible to beneficiary in advance	Real-time stock shown on dashboard
Order placement	In-person, queue-based	Digital, guided ordering with eligibility check
Delivery mode	Shop collection only	Home delivery or shop pickup
Tracking	None	Live order and agent tracking
Record of transaction	Paper register, easily disputed	Digital e-receipt and secure, timestamped proof of delivery
Fraud detection	Manual, after-the-fact audits	Automated flags on order patterns
Access control	Single shop-level authority	Hierarchical RBAC across three admin tiers

Fig. 5 illustrates two design-target metrics — average beneficiary wait time and an internal traceability score reflecting how completely each transaction can be reconstructed from digital records. Both are illustrative projections based on the elimination of in-person queuing and the introduction of end-to-end digital logging, not results measured from an operating deployment; Section VII specifies how they are to be measured.

Table II summarises the core functions implemented in each of the three portals, showing how responsibilities are distributed across the beneficiary, delivery, and administrative layers of the system.

Table II. Module-Wise Functional Summary

Module	Key Functions
User Portal	OTP login, live dashboard, guided ordering, delivery/pickup choice, multi-payment, order tracking, e-receipts
Delivery Agent Portal	Task list, route support, live status updates, customer contact, doorstep verification, proof of delivery
Admin Portal	RBAC dashboards, inventory & order management, delivery assignment, fraud alerts, reports & analytics

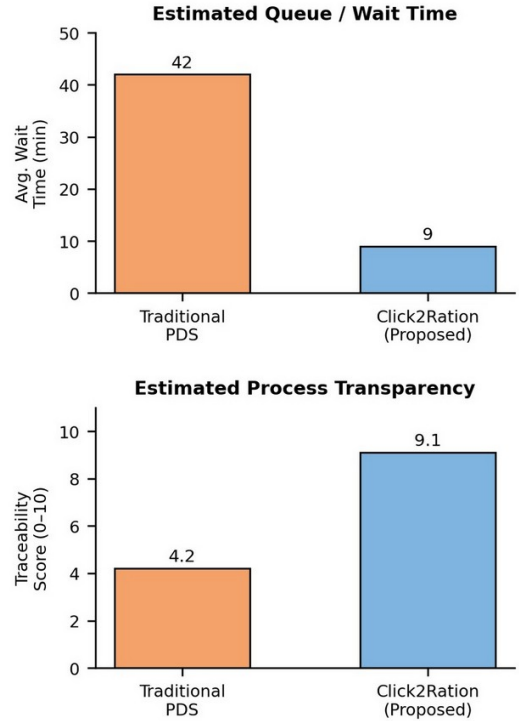


Figure 5. Projected wait-time and traceability improvement (design-stage estimates, pending pilot validation per Section VII).

VIII. SECURITY CONSIDERATIONS

Passwords are stored using bcrypt hashing, and all sessions use JWT tokens with configurable expiry and automatic logout after inactivity. API endpoints apply role-based authorisation checks, input validation, and rate limiting to resist brute-force and injection attempts. Every login, role change, and delivery verification is logged for audit purposes, and biometric or OTP verification at the doorstep is designed to prevent both unauthorised collection and false claims of non-delivery. Section VI details the specific threat model — device loss, network outages, and OTP/SMS spoofing — and the corresponding fallback and rate-limiting mechanisms that support these controls; the database tamper-evidence limitation and its proposed hash-chained mitigation are likewise discussed in Section VI.

All traffic between the three portals and the backend is intended to run over TLS, so that OTPs, session tokens, and beneficiary personal data are not exposed in transit even on shared or low-

trust rural network infrastructure. Geotags attached to proof-of-delivery records (Eq. 5) are validated server-side against the registered delivery address within a configurable tolerance radius, so that a delivery cannot be confirmed from a location inconsistent with the beneficiary's shop jurisdiction, adding a lightweight secondary check alongside OTP/biometric verification. Admin-side inputs are parameterised at the database layer to prevent SQL injection, and every RBAC-scoped query (Eq. 3) is filtered at the backend rather than the client, so a compromised or modified client application cannot widen an admin's visibility beyond their assigned district or shop.

IX. CONCLUSION AND FUTURE SCOPE

Click2Ration extends PDS digitisation beyond the ration-shop counter to a fully verified, trackable doorstep-delivery pipeline, combining OTP/biometric authentication, live tracking, hierarchical RBAC, and automated fraud flagging within a single platform. By ensuring that every order, delivery, and administrative action is verified and logged, the system aims to reduce beneficiary wait time, curb diversion of commodities, and give government authorities auditable visibility across the distribution chain. This revision adds an explicit threat model with defined fallback procedures for device loss, network outages, and OTP spoofing, a database schema formalised as an entity-relationship model, and a concrete pilot-evaluation protocol, directly addressing the absence of quantitative evaluation and edge-case handling identified in review. Future work includes executing the pilot study defined in Section VII, a native mobile application, AI-based demand forecasting for shop-level stock replenishment, a hash-chained or blockchain-backed audit ledger to upgrade the current secure-but-mutable log to a tamper-evident one, and deeper Aadhaar-linked biometric integration to further strengthen beneficiary identification.

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